



Reflection and E-Portfolio Report for Intelligent Agents Module

Student ID: 12697857

Module: Intelligent Agents

Total Word Count: 2746 Words

GitHub Repository URL: <https://fahadsaleh19.github.io/E-Portfolio/intelligent-agents.html>

Date: 20 October 2025

Reflective Of Intelligent Agents Module

Student ID: 12697857

Module: Intelligent Agents

Word Count: 1114 words

GitHub Repository URL: <https://fahadsaleh19.github.io/E-Portfolio/intelligent-agents.html>

Date: 20 October 2025

1. Introduction

The reflection is a review of my experience of learning during the Intelligent Agents module. It discusses the ways in which my perception of agent-based systems, Python programming, and software development changed during the course. I also learned to be more effective as a team player through GitHub, and my knowledge of the professional and ethical considerations in the autonomous system design was enhanced with the help of the module.

The reflection is organised in accordance with the model of Rolfe et al. (2001) What? So What? Now What? - to assess what I have learned, why it is relevant, and how I will apply such lessons in the new projects.

2. Description of Experience (What?)

Prior to this module, I only had a very general picture of Artificial Intelligence and little experience on agent-based systems (Russell and Norvig, 2021). The knowledge that I gained in the first unit is that Intelligent Agents are autonomous software programs that can react, reason and communicate within dynamic environments. Autonomy, proactivity, and social interaction were the concepts which were the basis of my knowledge.

During the initial units, I learned various agent architecture, including reactive, deliberative, and hybrid systems. I got to know of the support that each architecture offers to agents in decision making and problem solving. The use of small-scale examples in Python enabled me to relate theoretical concepts with real-life development. I also understood that programming logic is very essential and must be well-structured in order to make an agent act independently and respond to the change of environment.

One of the most important points of the module was the group project. The experience of developing an agent-based system in a team provided me with practical experience in working as a team and version control. GitHub was used to control the code, roles, and record the progress. My responsibilities were to maintain the repository, review commits and guarantee code uniformity (GitHub, Inc., 2025). This provided me with an experience of valuing the role of collaboration and discipline in software projects.

Subsequent courses broadened my understanding of the interaction of agents and coordination of activities by using fixed protocols. It was also interesting to me how I could get theoretical models of cooperation working into actual simulations with Python. **3. Introspection and Emotional Response (So What?)**

I experienced problems with the technical material at the beginning. These terms as agent communication languages, hybrid model and logical reasoning were tricky. The early project experiences with programming were discouraging because of syntax and logic errors. Nevertheless, these failures made me learn persistence and how to solve problems, which are essential skills of any developer.

My confidence increased as the weeks went by. The process of working with other team members on GitHub was more effective and I began to like debugging and refining our code. It was gratifying to see our system slowly form itself. It was just like the real world of software development where development is only progressed through cooperation, communication and consistency.

The emotional experience was the one that took me through the path of uncertainty to confidence. Based on the Reflective Cycle proposed by Gibbs, I realised that being frustrated

and reflecting is a learning process. Every difficulty, be it technical or co-operative, helped me to become a reflective practitioner.

The most influential was an ethical aspect of autonomous system design. I also got to know that intelligent agents need to be transparent and developers are responsible of their behavior (Wooldridge, 2020). The argument on accountability and fairness helped me to understand that design of technology is both ethical and technical Endeavour (Johnson, 2020). This knowledge transformed my perception on the idea of programming: it is not just a code but a system that can bring impact on people and society.

The application of GitHub during the module also enhanced my online professionalism. I was taught how to write my work in a way that would be well understood, how to maintain repositories and use version control in a responsible manner (Van Rossum and Drake, 2009). The platform promoted teamwork and gave me the feeling of being the owner of my contributions. Having a well organised repository reflected professional growth activities in the industry.

4. Lesson Learned and Future Application (Now What?)

The module has molded my professional and technical personality. It enabled me to evolve as a student who studied theory to a developer who can build and maintain intelligent systems as a responsible developer.

Technical Skills:

I acquired a good knowledge of the Python programming language and the concept of writing modular and reusable code. I learned how to combine agency behavior rationally and model

interactions. In the future, I will consider models like Mesa to build more advanced agent-based models.

Software Development: This stage involves the creation of effective software to ensure a company operates efficiently, Software Development: This is the phase that deals with development of efficient software to enable a firm work at its own pace.

I have discovered the value of version control, documentation and collaboration through GitHub. I have now the ability to manage the repositories, review code as well as handle conflicts effectively. Such skills will come in handy both academically and in work-related scenarios where teamwork and accountability is a must.

Professional Awareness: I am developing ethical awareness and professional awareness as well.

Ethical and Professional Awareness: The module improved my knowledge of professional responsibility. I also learned that smart systems may have an influence on the outcome of real-world relationships, and developers should pay attention to the ethical, social, and legal consequences of the design process. I will keep using the ethical reasoning to all the projects I will be working on.

Future Goals:

I have a number of goals to achieve in order to continue moving forward:

1. Keep on with Python by making open source contributions.
2. Use agile techniques of collaborative coding.
3. Further develop the understanding of agent-based simulation tools.

4. Reflect on a regular basis based on the model of Rolfe to assess learning and growth.

Such measures will assist me in maintaining technical and ethical proficiency.

5. Conclusion

Looking back at this module, I can see quite clearly the way my technical competence, teamwork, and professional mind have been shaped. I started with less knowledge of intelligent systems and now I have learned how to design code and evaluate intelligent systems using Python.

Project management via GitHub has helped me to understand that teamwork and organisation are the keys to success in software development. Simultaneously, the ethical aspect of the module informed me that the development of intelligent systems is a responsibility that should be transparent.

References:

GitHub, Inc. (2025) *Collaborating with Issues and Pull Requests*. Available at: <https://docs.github.com/en/issues> (Accessed: 18 October 2025).

Johnson, D.G. (2020) *Computer Ethics*. 5th ed. Pearson Education. Available at: <https://mrce.in/ebooks/Computer%20Ethics%204th%20Ed.pdf> (Accessed: 18 October 2025).

Russell, S. and Norvig, P. (2021) *Artificial Intelligence: A Modern Approach*. 4th ed. Harlow: Pearson Education. Available at: <https://aima.cs.berkeley.edu/> (Accessed: 18 October 2025).

Van Rossum, G. and Drake, F.L. (2009) *Python 3 Reference Manual*. Scotts Valley, CA: CreateSpace. Available at: <https://www.ecampus.com/python-3-reference-manual-van-rossum/bk/9781441412690> (Accessed: 18 October 2025).

Wooldridge, M. (2020) *An Introduction to MultiAgent Systems*. 3rd ed. Hoboken, NJ: John Wiley&Sons. Available at: https://uranos.ch/research/references/Wooldridge_2001/TLTK.pdf (Accessed: 18 October 2025).



E-Portfolio Report

Student ID: 12697857

Module: Intelligent Agents

Word Count: 1632 words

GitHub Repository URL: <https://fahadsaleh19.github.io/E-Portfolio/intelligent-agents.html>

Date: 20 October 2025

1. Introduction

This report describes how my e-Portfolio was developed and submitted during the Intelligent Agents module at the University of Essex. The e-portfolio hosted on GitHub contains the evidences of my knowledge development during all the twelve courses, including the knowledge of mechanics and the approaches to the real work in Python and the experience of working in a virtual collaborative team.

The module delved in the design, implementation, and evaluation of agent based systems that focused on architectures, communication protocols, hybrid methods, and ethical considerations. I have applied the e-Portfolio to subjectively examine how I have developed critical skills, including problem-solving, design of systems, expertise in coding, and responsible AI practice.

During the course period, I used a reflective team of Rolfe et al. (2001) and Kerb sphere of experiential learning (Kolb, 1984) to organise my development. Although the major part of the e-Portfolio is recorded with technical accomplishments, the combination of professional values, collaboration within a team, and ethics in the digital realm can be observed as well.

2. E-Portfolio Work description

This is my e-Portfolio based on GitHub and in the repository called Intelligent-Agents. It is a well-organised log of the weekly tasks, group discussions and work, and project work. All folders reflect the units of a module and hold markdown (.md) files and summaries of the learning results of every week, code snippets, screen captures, and notes of the evaluation.

The README.md file presented the general overview of the module and each unit had its own link. This made it easy and clear to navigate around. They were divided into thematic sections:

Unit 1 -4: essences of agents, and principles of their logic, and architectures.

Unit 5-8 unit: Agent communication, ontologies and natural language processing.

Unit 9 -12: Integrating deep learning with adaptive algorithms and intelligent agents.

Section by section entails theoretical summaries alongside the practical applications using Python. I performed some simulation activities in Jupiter Notebooks, which include agent prototypes involving reactive and deliberative agent, agent communication and message-passing protocols, as well as experiments in ontology-based decision systems.

There were screenshots, annotated code snippets and comments by peers, which were also used to show teamwork and contribution within group work. The e-Portfolio is therefore not just a document of learning but it is also a professional demonstration of cooperative software development.

3. Key Learning Outcomes

3.1 Python and Comparable Development

I developed technical capability, which was among the most important outcomes of this module. Using the introductory lessons, I was able to understand the distinction between intelligent agents and traditional programs, having autonomy, reactivity, proactivity, and the ability to socialize (Wooldridge, 2021). A translation of these theoretical concepts into practice code enhanced my knowledge.

In Python, I wrote simple reflex agents, model-based agents followed by goal-based designs on the models. Unit 3 involved me learning about Belief-Desire-Intention (BDI) architecture that enabled agents to strike a balance between dynamic decision-making and long-term goals. Use of Python libraries e.g. spade and Mesa helped me in simulating multi-agent environment, implement communication protocols, and visualization of agent interactions.

This application practice enabled my mental software engineering attitude - especially debugging, modularity, and algorithm reasoning. In accordance with Russell and Norvig (2022), intelligent agents need to bring together perception, reasoning, and action together. I created agents able to adapt to the evolving environments through trial and error, which further supports the importance of the module with regard to uncertainty and risk management.

The use of Github teamwork helped me to get more familiar with the real working processes. I created the agile development cycle with the aid of commits, pull requests, and issue tracking.

This reflected the practice of professional engineering, which Pressman and Maxim (2020) recommend to make sure that my coding was transparent and reproducible.

3.2 Virtual Collaboration and Teamwork

The teamwork task aspect played a crucial role in establishing communication and coordination capabilities. We were in a distributed virtual team in which we developed an agent-based resource allocation system. The project members had particular roles assigned to them - project coordinator, project developer, ontology designer, and project documenter. My most common contributions were Python and version control at GitHub.

GitHub, Microsoft Teams and shared Google Docs were also part of our working tools, where we proposed the design and regularly updated on its progress. This is whereby technical and interpersonal skills were nurtured as is in the theory of Belbin (2010) on team roles that underlines equilibrium in skills in effective teams.

Sometimes there was conflict on a basis of version blending and role duplication but we used reflective framework of Gibbs (1988) to solve the problems of communication constructively. We have been taught to be able to record the changes made well, allocate tasks openly, and make use of pull requests to enable us to work together effectively. This experience enhanced my knowledge on the field of professional software development teamwork and virtual project management.

The possible finished product - a prototype hybrid agent system not only displayed technical integrity, but also the non-technical skills of negotiation, leadership, and adaptability. Our design has blended reactive and deliberative layers demonstrating that hi-tech architecture can provide a trade-off between performance and flexibility (Ferber, 1999).

Each person possesses unique principles to adhere to and standards to consider.

3.3 Professional and Ethical Awareness

Every individual has his or her own principles to follow and norm in consideration.

All through the module, ethical, legal, and social considerations were incorporated. We analysed Unit 10, where the author discussed how bias or privacy can be generated by using AI systems such as agents without being responsible in their creation. This was more applicable in the case of simulated decision data when we were dealing with group project.

In accordance with the University of Essex Academic Integrity Policy (2023), all the coding and documentation were made ethically in regard to authorship and data transparency. We similarly considered the principles of AI ethics provided by Floridi and Cowls (2019) beneficence, non-maleficence, justice, and explicability, to make sure that our system design, based on an agent, was also just and responsible.

With the help of such a moral prism, I got to know that the creation of intelligent agents is not merely a technical endeavor. It deals with ethical thinking and the consideration of social influence. These views equipped me with an understanding of the professional practice in both AI and software development wherein the adherence to the standards, such as Ethically Aligned Design (2021) of the IEEE is mandatory.

4. Reflection and critical Thinking

I will take the example of formulating inter-agent communication with ontologies where I observed the reflection of miscommunication in human beings. This further enhanced my sense of the semantic accuracy and design clarity. In a similar way, further courses studying adaptive-based algorithms showed me the connection between agents and machine learning, showing the way of generating intelligence through feedback and learning systems.

My professional identity was also affected by the experience. A software development process now appears to me as a team-building and morally right activity, not as a separate technical one. According to Kolb (1984), learners create new knowledge when they can change their experiences by reflecting through the process of learning which is facilitated in this module by documentation and evaluation in e-portfolio on the team level.

5. Future Uses and Future Development

The Intelligent Agents module not only supplied me with theoretical but also practical abilities which can be used later on in AI and software development. I intend to develop them in further study and cultivate my profession.

My second objective is to model an agent-based energy manager simulation based on the Mesa library of Python and reinforcement learning to perform adaptive control. This will integrate the ideas of Units 9-11 adaptive algorithms, communication, and deep learning in solving the issue of sustainability.

In order to enhance my employability even further, I have established the following SMART objectives:

- **Specific:** Produce and present a GitHub project that is a representation of a hybrid agent system respectful of ethical concerns.
- **Measurable:** Adapt and test communications specifications and decision making units by the month of March 2026.
- **Achievable:** Spend 4 hours per week on Python coding as an agent, and documenting the project.
- **Relevant:** Make the project match the Industry 4.0 applications, specifically, in the field of IoT and automation.
- **Time limit:** hand over at the expiry of the following academic term.

Besides this, I intend to donate to open-source communities that are researching on intelligent agents. Engaging in GitHub discussions will enable me to work and cooperate with people around the world and put the lessons into practice.

In the long term, I want to apply to the AI development or automation design that will be aimed at human collaboration, but not the elimination. Because, as Russell (2019) puts it, the implementation of AI in accordance with the human values is the key to the sustainable innovation, I have implemented this core concept into my professional vision.

6. Conclusion

The creation of the e-Portfolio on the Intelligent Agents module was a professional and an academic achievement. It helped to solidify my knowledge of agent-based systems, improve my knowledge of Python and strengthened my appreciation of teamwork, ethics and lifelong learning.

By using the collaboration with GitHub and critical thinking about theoretical frameworks, I got acquainted with the idea of how smart agents may be designed to act responsibly in the environment of uncertainty. The process coupled this concept, that successful AI development requires not just skills in coding, but also, and more importantly, reflection, communication, and ethical accountability. This experience has equipped me to use such competencies in practice situations, when intelligent systems need to serve the needs of people, be fair, and transparent.

References:

Belbin, R. M. (2010). *Team Roles at Work* (2nd ed.). Oxford: Butterworth-Heinemann. Available at: <https://www.taylorfrancis.com/books/mono/10.4324/9780080963242/team-roles-work-meredith-belbin> Taylor & Francis+2PagePlace+2 (Accessed: 18 October 2025).

Ferber, J. (1999). *Multi-Agent Systems: An Introduction to Distributed Artificial Intelligence*. Harlow: Addison-Wesley. Available at: <https://archive.org/details/multiagentsystem0000ferb> Internet Archive+2BibSonomy+2 (Accessed: 18 October 2025).

Floridi, L. & Cowls, J. (2019). “A Unified Framework of Five Principles for AI in Society.” *Harvard Data Science Review*, 1(1). Available at: <https://doi.org/10.1162/99608f92.8cd550d1> Harvard Data Science Review (Accessed: 18 October 2025).

IEEE Standards Association. (2021). *Ethically Aligned Design: A Vision for Prioritizing Human Well-being with Autonomous and Intelligent Systems*. Available at: https://standards.ieee.org/wp-content/uploads/import/documents/other/ead_v2.pdf IEEE Standards Association (Accessed: 18 October 2025).

Kolb, D. A. (1984). *Experiential Learning: Experience as the Source of Learning and Development*. Englewood Cliffs, NJ: Prentice Hall. Available from: https://www.researchgate.net/publication/235701029_Experiential_Learning_Experience_As_The_Source_Of_Learning_And_Development (Accessed: 18 October 2025).